

Class I, II, III Forklifts and the Right Lithium Battery Configuration

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Ask three different warehouse managers what "the right forklift battery" means, and you'll likely get three different answers — because a Class III pallet jack, a Class II reach truck, and a Class I counterbalanced rider forklift don't just look different, they run on entirely different electrical platforms. Voltage, amp-hour capacity, connector type, and even physical footprint all shift depending on which class of truck you're powering.

This distinction matters more than ever now that lithium iron phosphate (LiFePO4) has become the default upgrade path away from lead-acid. Unlike lead-acid, where a facility might improvise with whatever battery physically fits, a lithium pack is a precision component — matched to the truck's electrical system, its compartment dimensions, and its duty cycle. Ordering the wrong voltage doesn't just underperform; in most cases, it simply won't power the truck at all.

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This guide walks through how OSHA defines Class I, II, and III forklifts, what voltage and capacity each class typically demands, and which lithium configuration actually fits — using real specifications so you can plan a purchase instead of guessing.

How OSHA Defines Class I, II, and III Forklifts

The Occupational Safety and Health Administration groups powered industrial trucks into seven classes, and the first three cover every electric-motor-driven truck on a typical warehouse floor. According to **OSHA's own classification framework**, Class I covers electric motor rider trucks, Class II covers electric motor narrow aisle trucks, and Class III covers electric motor hand trucks or hand/rider trucks. That three-way split is also the foundation the **ANSI/ITSDF B56 series of safety standards** builds on when defining design, operation, and maintenance requirements for each truck type.

OSHA Class	Common Name	Typical Trucks	Operator Position
Class I	Electric Motor Rider Trucks	Counterbalanced sit-down and stand-up forklifts	Riding
Class II	Electric Motor Narrow Aisle Trucks	Reach trucks, order pickers, turret trucks	Riding or standing
Class III	Electric Motor Hand Trucks / Hand-Rider Trucks	Pallet jacks, walkie stackers, low-lift trucks	Walking or riding

The classification itself doesn't specify a battery voltage — that's determined by the individual truck manufacturer and model. But in practice, decades of OEM engineering conventions have settled into predictable voltage bands for each class, which is exactly what a lithium replacement battery has to match.

Class III: Electric Motor Hand Trucks

Class III trucks — pallet jacks, walkie stackers, and low-lift equipment — are the entry point of the electric fleet. They're lighter, simpler machines, and their electrical systems reflect that: most run on **24V**, occasionally stepping up to 36V for higher-capacity walkie stackers or hand/rider models.

Typical Class III requirements:

- **Voltage:** 24V (36V on some larger walkie/rider models)
- **Capacity range:** 150–300Ah is common for standard pallet trucks
- **Physical profile:** Low, flat compartments — battery height is often the tightest constraint, not weight
- **Duty pattern:** Frequent stop-start cycles, shorter runtimes per shift, ideal candidates for opportunity charging

The **BlitzNXT 24V 230Ah LiFePO4 forklift battery** is built specifically for this class. At 25.6V nominal and 5.9kWh of usable energy, it delivers a 115A continuous discharge current with a 175A peak — enough to keep a pallet truck's hydraulic lift and drive motor responsive through a full shift without voltage sag. Its low-profile 662 × 650 × 195mm case and roughly 105kg weight are sized to slide into the same compartment as the lead-acid unit it replaces, and its built-in charger port and circular battery-level indicator mean an operator can read state of charge without a phone app or external display.



Class II: Electric Motor Narrow Aisle Trucks

Class II trucks operate in tighter spaces at greater lift heights — reach trucks, order pickers, and turret trucks working narrow-aisle high-density racking. These trucks generally need more sustained power than Class III equipment but don't carry the mass of a full counterbalanced rider truck, so **36V and 48V** platforms dominate this class.

Typical Class II requirements:

- **Voltage:** 36V or 48V, depending on lift height and mast configuration
- **Capacity range:** Mid-to-high Ah, often 400Ah and above for multi-shift operations
- **Physical profile:** Compartment dimensions vary significantly by OEM and mast type — always confirm against the nameplate before ordering
- **Duty pattern:** Extended lift cycles at height place sustained load on the pack; voltage stability under load matters more here than in Class III

For 36V platforms, the **BlitzNXT 36V 920Ah LiFePO4 forklift battery** is built for reach trucks and narrow-aisle equipment running extended multi-shift schedules, where high total energy storage reduces how often the truck needs to return to a charging station. Where a facility's narrow-aisle fleet runs on 48V systems instead, the same 51.2V platform used in Class I counterbalanced trucks (detailed below) applies equally well.



Class I: Electric Motor Rider Trucks (Counterbalanced)

Class I trucks are the workhorses most people picture when they hear "forklift" — sit-down and stand-up counterbalanced riders handling palletized freight throughout a warehouse or distribution center. Because these trucks carry the heaviest loads and often run continuously across multiple shifts, they demand the widest voltage range of any class: **36V, 48V, and 80V** are all common, depending on load capacity and lift height.

Typical Class I requirements:

- **Voltage:** 48V is the most common mid-size platform; 80V is standard on heavier-capacity and high-lift models
- **Capacity range:** 400–900+ Ah, sized to cover a full shift or more between charges
- **Physical profile:** The battery itself typically serves as part of the truck's counterbalance, so weight and dimensions must match the OEM nameplate closely — not just fit the compartment
- **Duty pattern:** Sustained high-current draw under load, frequent opportunity charging during breaks in multi-shift facilities

Two BlitzNXT models illustrate the range within Class I:

The **BlitzNXT 48V 460Ah LiFePO4 forklift battery** is built at 51.2V nominal, delivering 23.6kWh of usable energy with a 100A continuous discharge current (350A peak) — enough to run a standard 48V

counterbalanced or sit-down rider truck through a full shift without a charging interruption. At 32.7 × 22.0 × 16.0in and roughly 507lbs, it's sized as a direct swap into standard 48V compartments, and its REMA320/Anderson350 connector interface and IP67-rated case are built for daily plug-cycle use on a warehouse floor. For higher-capacity 48V fleets, the **BlitzNXT 48V 608Ah LiFePO4 forklift battery** extends that same platform to longer multi-shift runtimes.

For heavier-duty Class I trucks — large-capacity counterbalanced and reach forklifts operating at higher lift heights — the **BlitzNXT 80V 460Ah LiFePO4 forklift battery** delivers 36.8kWh of usable energy at the same 100A continuous / 350A peak discharge rating, but at double the voltage. Higher voltage means the truck's motor draws less current for the same power output, which translates to less resistive heating, smoother response on heavy lifts, and better efficiency across long shifts. At roughly 595lbs and 39.0 × 24.0 × 16.9in, it's sized as a like-for-like replacement for 80V lead-acid packs without structural modification to the truck.



Side-by-Side: Matching Class to Configuration

Truck Class	Typical Voltage	Typical Ah Range	Example BlitzNXT Model	Usable Energy	Continuous / Peak Discharge
Class III (Hand/Pallet Trucks)	24V	150–300Ah	24V 230Ah	5.9kWh	115A / 175A
Class II (Narrow Aisle)	36V–48V	400–900+ Ah	36V 920Ah	~35kWh	Model-dependent
Class I (Counterbalanced, standard)	48V	400–650Ah	48V 460Ah	23.6kWh	100A / 350A
Class I (Counterbalanced, heavy-duty)	80V	400–500Ah	80V 460Ah	36.8kWh	100A / 350A

This table is a starting reference, not a substitute for the truck's own nameplate. Two forklifts in the same OSHA class, even from the same manufacturer, can specify different voltages depending on load capacity and mast configuration — which is why voltage confirmation always has to come from the truck itself, not the class alone.

What to Check Before You Order

Getting the class right narrows the field, but four additional checks determine whether a specific lithium battery is actually the correct replacement:

- **Voltage match is non-negotiable.** A 48V truck cannot run — safely or at all — on a 36V or 80V pack. This is the first and least flexible specification.
- **Compartment dimensions, not just voltage.** Confirm length, width, and height against the physical battery, especially on Class II and III trucks where compartments are often tightly sized.
- **Weight and counterbalance.** On Class I counterbalanced trucks, the battery's mass contributes to the truck's stability rating. A lithium pack that's substantially lighter than the lead-acid original can shift the truck's load center — worth reviewing before a fleet-wide swap. For a deeper look at this specific issue, see our guide on how lithium forklift battery weight changes counterbalance.
- **Connector type and BMS communication.** REMA and Anderson-style connectors dominate the market, but pin configuration and current rating still need to match. If your fleet reports battery data into a warehouse management system, confirm the pack's COM port supports your telematics setup.

Every one of these fields — voltage, Ah, and compartment size — is printed directly on the existing battery's nameplate. If you're staring at a lead-acid nameplate and aren't sure how to translate it into a lithium spec, our guide on how to read a forklift battery nameplate walks through every field with a worked example.

Why the Class Distinction Matters More With Lithium

Lead-acid tolerance for a "close enough" fit made voltage mismatches easier to paper over in the past — an undersized battery would just underperform and need more frequent watering and equalization. LiFePO4 removes that slack. A correctly sized lithium pack, by contrast, is a maintenance-free component for the life of the truck: no watering, no acid handling, no equalization charging, and — across a Class I, II, or III fleet alike — a flat voltage curve that keeps lift speed and drive performance consistent from a full charge down to empty. For a full breakdown of what that maintenance reduction actually looks like day to day, see what lithium forklift batteries eliminate from your daily maintenance checklist. Facilities converting an entire fleet across multiple truck classes also typically reclaim the square footage previously reserved for a dedicated lead-acid charging and watering room — a shift covered in our guide on how lithium forklift batteries free up warehouse floor space.

Safety Standards That Apply Across All Three Classes

Regardless of class or voltage, any lithium forklift battery entering a North American facility should be evaluated against two layers of standards. The truck itself falls under ANSI/ITSDF B56.1, the safety standard covering low lift and high lift powered industrial trucks controlled by a riding or walking operator on compacted, improved surfaces — the design and operating standard referenced by OSHA's own powered industrial truck regulations. The battery pack itself is a separate component, and its safety testing is generally evaluated against UL's standard for batteries used in electric vehicles, which covers electrical, mechanical, and environmental abuse testing for the

pack and its subassemblies — including battery assemblies used in industrial trucks. Asking a supplier which standards their specific voltage platform has been evaluated against is a reasonable, and increasingly standard, procurement question.

Runtime Planning: Matching Ah to Shift Length

Voltage gets a truck running at all; amp-hour capacity determines whether it keeps running long enough to matter. This is where fleet managers most often either overspend on capacity they don't need or undersize a battery that ends up on the charger mid-shift.

A rough planning approach:

- **Single-shift, light-duty Class III operations** (occasional pallet moves, short travel distances) can often run comfortably on the lower end of the Ah range for their voltage — a 24V 230Ah pack, for example, covers a full standard shift with room to spare for most pallet truck duty cycles.
- **Multi-shift Class I and II operations** should size capacity to the full working day, not a single shift, if opportunity charging during breaks isn't part of the workflow. LiFePO4's tolerance for partial-cycle charging makes short top-ups during scheduled breaks a practical way to extend a smaller pack's effective runtime — but that only works if the facility actually has charging infrastructure positioned where operators take breaks.
- **Heavy-duty Class I trucks running continuous three-shift schedules** typically justify the higher-capacity end of the range (600Ah+ at 48V, or the full 460Ah at 80V) specifically to avoid mid-shift charging interruptions on trucks that are rarely idle long enough for a meaningful opportunity charge.

One advantage lithium has over lead-acid in this planning exercise: because LiFePO4 delivers a flat voltage curve and supports charging to any state of charge without the equalization requirements lead-acid imposes, a facility can size capacity closer to actual daily energy consumption rather than padding for the voltage sag and reduced usable depth of discharge that lead-acid batteries require.

Getting the Configuration Right the First Time

The fastest way to avoid an ordering mistake is to start from the truck, not the battery. Pull the nameplate voltage, confirm the OSHA class and compartment dimensions, and cross-reference against the Ah capacity your shift length actually requires — not the highest number available. A 24V Class III pallet truck, a 48V Class I counterbalanced rider, and an 80V heavy-duty counterbalanced truck are three different electrical worlds, and no single lithium SKU bridges all three.

Power Every Class in Your Fleet With BlitzNXT

BlitzNXT builds LiFePO4 forklift batteries across 24V, 36V, 48V, and 80V platforms — engineered as drop-in replacements for Class I, II, and III trucks alike, with Grade-A EVE cells, IP67 sealing, and a 10-year warranty behind every pack. Whether you're replacing a single pallet jack battery or converting a mixed-class fleet, our team can confirm the exact voltage, capacity, and connector your nameplate calls for before

you order. Browse the full forklift battery lineup or contact our team for a fleet-specific recommendation.

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